
2-shot prompt

<s>[INST] «SYS» Given the description after "Description:", write a true statement about all boxes and their contents to the description after "Statement:". «/SYS»

Description: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map. [/INST] Statement: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map. </s>

<s>[INST] Description: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map. Remove the car from Box 0. Remove the paper and the string from Box 3. Put the plane into Box 0. Move the map from Box 6 to Box 2. Remove the bill from Box 4. Put the coat into Box 3. [/INST] Statement: Box 0 contains the plane, Box 1 contains the cross, Box 2 contains the bag and the machine and the map, Box 3 contains the coat, Box 4 contains nothing, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle. </s>

<s>[INST] Description: description

Table 5: Chat-formatted prompt with 2-shot in-context demonstrations.

In our experiments, we found that the models struggled to adhere to the output format specified via the few-shot prompt examples. Luckily, the expected output can be described precisely by the regular expression shown in Figure 4. We used the outlines library (Willard & Louf, 2023) which supports regex-based constrained decoding. We found a significant improvement with constrained decoding. For e.g., the performance of the Llama 2 70B model went up from 54.95 to 62.13 with constrained decoding.

A.3 Detailed Results

Table 6 presents the detailed results of all the models evaluated in this work. The results are categorized by the number of operations affecting the entity of interest.

Model	Performance split by number of operations								
	Overall (5012)	0 (1303)	1 (1410)	2 (1083)	3 (651)	4 (288)	5 (142)	6 (106)	7 (29)
Random	21.08	41.06	17.85	12.70	11.87	10.58	9.16	7.51	12.59
Llama 2-7B	21.31	65.54	4.96	7.94	4.30	5.56	4.93	5.66	3.45
Llama 2-7B Chat	41.28	87.95	35.53	24.38	14.29	12.15	14.79	7.55	3.45
Llama 2-13B	33.28	77.05	25.18	15.97	12.75	8.68	8.45	10.38	17.24
Llama 2-13B Chat	38.05	77.51	36.67	19.94	16.28	9.03	11.27	11.32	13.79
Llama-2 70B	62.13	99.00	71.91	43.67	31.18	21.88	25.35	25.47	27.59
Llama-2 70B Chat	63.43	93.25	75.67	47.00	37.33	26.39	22.54	24.53	37.93
Code Llama 7B	31.94	90.87	13.69	12.28	7.99	6.60	4.93	8.49	13.79
Code Llama 7B Instruct	41.34	96.16	36.03	18.19	12.29	6.60	5.63	3.77	10.34
Code Llama 13B	54.79	95.70	60.78	32.87	25.35	20.14	21.83	26.42	13.79
Code Llama 13B Instruct	58.14	97.77	71.13	35.83	23.35	17.01	14.79	19.81	20.69
Code Llama 34B	58.26	94.70	71.28	33.89	31.18	21.53	14.79	19.81	24.14
Code Llama 34B Instruct	61.47	95.09	77.09	39.98	31.03	22.57	19.01	20.75	20.69
Code Llama 70B	69.77	99.39	76.60	54.20	47.31	40.97	36.62	43.40	37.93
Code Llama 70B Instruct	73.66	98.70	83.40	59.65	51.92	46.18	40.85	39.62	44.83
Llemma 7B	34.12	84.65	26.81	13.20	8.29	4.51	6.34	9.43	0.00
Llemma 34B	60.18	95.24	75.60	36.10	31.64	21.53	14.79	21.70	20.69
DeepSeek 7B	23.46	71.83	8.44	7.29	4.61	1.04	2.11	3.77	6.90
DeepSeek 7B Chat	32.24	70.30	27.30	17.27	11.21	11.81	9.15	6.60	3.45
DeepSeek 7B Coder Base	53.67	97.16	57.52	33.61	23.96	19.10	17.61	8.49	13.79
DeepSeek 7B Coder Instruct	48.76	91.71	54.54	26.41	18.13	13.89	11.27	17.92	3.45
DeepSeek 7B Math Base	56.40	98.93	60.28	35.73	26.73	24.65	16.20	23.58	27.59
Gemma-7B	42.66	95.78	45.82	14.04	8.60	5.56	7.75	7.55	3.45
Gemma-7B Instruct	47.79	93.25	55.96	21.98	15.51	7.99	8.45	11.32	17.24
CodeGemma-7B	46.09	91.33	53.76	21.70	13.67	6.94	5.63	8.49	3.45
CodeGemma-7B Instruct	53.49	93.48	69.57	26.22	19.82	12.50	12.68	11.32	10.34
Mistral 7B	45.67	96.01	50.28	18.56	11.37	8.68	12.68	7.55	10.34
OpenMathMistral	38.25	90.94	34.40	14.22	8.76	5.90	4.93	5.66	20.69
LLama 7B	28.67	97.93	0.71	10.53	2.92	3.82	3.52	0.94	3.45
Float 7B Instruct	27.55	89.33	1.56	12.37	5.07	5.21	2.82	4.72	13.79

Table 6: Entity Tracking performance of models categorized by the number of operations affecting the entity of interest. The count of test set instances with the number of operations affecting an entity is indicated in parentheses below the corresponding column title.